In-Class Practice: K-Means Clustering

Selecting k, Normalization, Training, and Predicting a New Point

Objective

Use K-Means clustering to group customers using **Age** and **Income($)**, determine a suitable number of clusters using the Elbow Method, and assign a new customer to one of the learned clusters.

Input Files

File	Columns	Purpose
income.csv	Name, Age, Income($)	Training data
new_point.csv	Name, Age, Income($)	New point for prediction

Important: Use only Age and Income($) for clustering. Name is an identifier and must not be used to calculate distances.

Tasks

1. Load and inspect the data

Load income.csv and confirm the attributes used for clustering.

2. Normalize the clustering attributes

Apply Min-Max normalization to Age and Income($). Fit the normalization model/scaler using the training data only.

3. Determine an appropriate value of k

Run K-Means for k = 1 to 9. Record WCSS/inertia (or equivalent within-cluster distance), create an Elbow plot, and state the k you choose with a short explanation.

4. Train the final K-Means model

Train K-Means using the selected k and normalized training data. Show the resulting clusters and centroids.

5. Predict the new point

Load new_point(1).csv. Normalize it using the SAME training normalization model/scaler, then assign it to a cluster using the trained K-Means model.

6. Interpret the result

Report the predicted cluster number and briefly describe that cluster using its centroid or member records.

Required Evidence in the PDF Answer

- Elbow plot for k = 1 to 9.
- Selected k with a short justification.
- Final clustering result showing the clusters.
- Cluster centroids or brief cluster descriptions.
- The new point and its predicted cluster.
- Confirmation that the new point used the same training scaler/normalization model.

Submission

Submit exactly two files:

1. One implementation file: either .ipynb OR .xml
2. One PDF answer file containing the required evidence and short answers.

Suggested filenames: StudentID_Name_KMeans.ipynb/xml and StudentID_Name_KMeans_Answer.pdf

Important Notes

- Cluster labels (e.g., 0, 1, 2) are arbitrary IDs, not rankings.
- For .xml, the process must clearly show preprocessing, K-Means training, and application of the trained preprocessing/model to the new point.

Check before submission

Item	Criteria
Preparation	Correct features + normalization
Elbow	k = 1..9, plot, reasonable k
Final model	Correct K-Means result
Prediction	Same scaler + correct application
PDF	Clear evidence + interpretation